

### Module A: Least-squares view (12.1) + minimum-norm exact solution (12.3)

**Big picture** All these formulas come from *projection geometry*: least squares projects  $b$  onto  $\mathcal{R}(A)$  to **find the best solution among all imperfect ones**; minimum-norm solutions pick the solution in  $\mathcal{R}(A^\top)$  (orthogonal to  $\mathcal{N}(A)$ ) to **find the simplest solution among the equivalently effective ones**.

#### 12.1 Least squares for $Ax \approx b$ .

Given  $A \in \mathbb{R}^{m \times n}$ ,  $b \in \mathbb{R}^m$ :

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2.$$

**Normal equations.** Any minimizer satisfies

$$A^\top(Ax - b) = 0 \quad \Leftrightarrow \quad A^\top Ax = A^\top b.$$

**Full column rank** ( $m \geq n$ ,  $\text{rank}(A) = n$ ). Then  $A^\top A \succ 0$  and the minimizer is unique:

$$x_{\text{LS}} = (A^\top A)^{-1} A^\top b.$$

**Geometric intuition.** The residual  $r = b - Ax_{\text{LS}}$  is orthogonal to  $\mathcal{R}(A)$ , i.e.,  $r \perp \mathcal{R}(A)$ .

#### 12.3 Minimum-norm solution of $Ax = b$ (underdetermined).

Assume  $A \in \mathbb{R}^{m \times n}$ ,  $m \leq n$ ,  $\text{rank}(A) = m$  (full row rank) and  $b \in \mathbb{R}^m$ . There are infinitely many  $x$  with  $Ax = b$ ; we choose the closest to the origin:

$$\min_{x \in \mathbb{R}^n} \|x\| \quad \text{s.t.} \quad Ax = b.$$

**Key theorem (minimum-norm formula).** The unique minimizer is

$$x^* = A^\top (AA^\top)^{-1} b.$$

**Intuition.** Any solution can be decomposed as  $x = x_{\mathcal{R}(A^\top)} + x_{\mathcal{N}(A)}$ . The constraint  $Ax = b$  fixes the  $\mathcal{R}(A^\top)$  component; the minimum norm sets the nullspace component to 0, hence  $x^* \in \mathcal{R}(A^\top)$ .

#### ! Remarks

- $\mathcal{R}(A^\top)$  could be understood as the **effective domain** of  $A$ .
- Two “projections”:**  $A(A^\top A)^{-1} A^\top$  is the projector onto  $\mathcal{R}(A)$  (when  $\text{rank}(A) = n$ );  $A^\top (AA^\top)^{-1} A$  is the projector onto  $\mathcal{R}(A^\top)$  (when  $\text{rank}(A) = m$ ).

#### 📖 Practice: Textbook 12.18 (projection onto $\mathcal{R}(A)$ )

Let  $A \in \mathbb{R}^{m \times n}$ ,  $b \in \mathbb{R}^m$ ,  $m \geq n$ , and  $\text{rank}(A) = n$ . Consider

$$\frac{1}{2} x^\top x - x^\top b \quad \text{subject to} \quad x \in \mathcal{R}(A),$$

where  $\mathcal{R}(A)$  denotes the range of  $A$ . Derive an expression for the global minimizer in terms of  $A$  and  $b$ .



## Module B: Kaczmarz's algorithm = cyclic row projections for Min Norm Solution (12.4)

We want to solve a consistent system  $Ax = b$  with  $A \in \mathbb{R}^{m \times n}$ ,  $m \leq n$ ,  $\text{rank}(A) = m$ . Write the rows of  $A$  as  $a_1^\top, \dots, a_m^\top$  and  $b = (b_1, \dots, b_m)^\top$ .

**Algorithm (cyclic row updates).** Choose  $0 < \mu < 2$  and an initial point  $x^{(0)}$ . For  $k = 0, 1, 2, \dots$ , pick the row index

$$j = R(k) + 1 \quad (\text{i.e., } j = (k \bmod m) + 1),$$

and update

$$x^{(k+1)} = x^{(k)} + \underbrace{\mu \frac{b_j - a_j^\top x^{(k)}}{a_j^\top a_j}}_{\text{project direction orthogonally towards } b_j = a_j^\top x^{(k)}} a_j.$$

**Geometric intuition.** The equation  $a_j^\top x = b_j$  defines a hyperplane where this entry of data constraint is satisfied / loss is 0. When  $\mu = 1$ , the update is the *orthogonal projection* of  $x^{(k)}$  onto that hyperplane. Cyclic Kaczmarz repeatedly projects onto each row-hyperplane.

**Convergence to Minimum Norm Solution.** If  $x^{(0)} = 0$ , then  $x^{(k)}$  converges to the *minimum-norm solution*

$$x^\star = A^\top (AA^\top)^{-1} b.$$

More generally, if  $x^{(0)} \neq 0$ , it converges to the point in  $\{x : Ax = b\}$  closest to  $x^{(0)}$ .



### Remarks

- $\mu$  is a **relaxation** parameter:  $\mu = 1$  is pure projection;  $0 < \mu < 2$  is required for the standard convergence proof.
- The method is **row-scaled**: the factor  $1/(a_j^\top a_j)$  normalizes the step by the row norm.



### Practice: Textbook 12.28 (projection property when $\mu = 1$ )

Show that in Kaczmarz's algorithm, if  $\mu = 1$ , then

$$b_{R(k)+1} - a_{R(k)+1}^\top x^{(k+1)} = 0 \quad \text{for each } k.$$